Huawei Access Node KR-ACC-15 Hardware/Environmental Fault Troubleshooting SOP

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2866

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for identifying and resolving hardware/environmental faults on Huawei Access Node KR-ACC-15. The failure domain, characterized by ALARM_2866 (ASIC memory parity errors, line card crashes, power supply redundancy loss), necessitates a comprehensive approach to isolate the root cause. The Huawei Access Node, as a critical component of the access network, relies heavily on the Line Card (LC) and Power Supply (PS) redundancy to ensure continuous service delivery. However, hardware failures or environmental stress can compromise this redundancy, leading to critical alarms and service disruptions. This SOP aims to guide the network engineer through a structured diagnostic process, leveraging Huawei's CLI dialect to identify and rectify potential issues related to the ASIC memory parity errors, line card crashes, or power supply redundancy loss.

Pre-requisite Checks

- Verify the site location and Node ID to ensure accuracy.
- Confirm the redundant link status and ensure it is functioning correctly.
- Dispatch remote hands to KR Puram Access for physical assistance, if required.
- Obtain necessary permissions and access to the Node's console.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display board all, Expected Output: Board information: Slot 1: Line Card, Status: Online, Description: This command displays the board information, including the slot number, type, and status. It helps to confirm the presence and status of the Line Card.

Command: display power all, Expected Output: Power Supply information: Slot 2: Power Supply, Status: Online, Description: This command displays the power supply information, including the slot number, type, and status. It helps to confirm the presence and status of the power supply.
```

```
Command: diag snmp error-record, Expected Output: Error record: ASIC memory parity error occurred at Slot 1, Time: 2023-02-20 14:30:00, Description: This command displays the error record, including the type of error, slot number, and timestamp. It helps to identify the source of the ASIC memory parity error.
```

Resolution Steps

Step 1: Step: Replace the faulty Line Card, Command: replace board slot 1 line-card, Description: Replace the faulty Line Card with a new one to rectify the hardware issue. Ensure to configure the new Line Card with the correct settings and parameters.

Step 2: Step: Verify the Power Supply status, Command: display power all, Description: Verify the Power Supply status to ensure it is functioning correctly. If the Power Supply is faulty, replace it with a new one.

Step 3: Step: Reset the Node, Command: reset system, Description: Reset the Node to clear any existing issues or configurations. Ensure to configure the Node with the correct settings and parameters after the reset.

Post-Check Verification

```
Command: display board all, Expected Output: Board information: Slot 1: Line Card, Status: Online, Description: Verify the Line Card status after replacement.

Command: display power all, Expected Output: Power Supply information: Slot 2: Power Supply, Status: Online, Description: Verify the Power Supply status after replacement.

Command: diag snmp error-record, Expected Output: No error record found, Description: Verify that no error record is found after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: system restore, Description: Restore the original configuration to revert any changes made during the resolution steps.

- Step: Replace the original hardware, Command: replace board slot 1 line-card, Description: Replace the original hardware with the faulty one to revert any changes made during the resolution steps.